

Iodine-125 Radiotherapy for Choroidal Melanoma

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The objective was to evaluate the effect of the gender, size, and tumor location at the time of the diagnosis on the regression response of choroidal melanoma following plaque radiotherapy treatment. The paper is a longitudinal prospective study of 28 patients diagnosed with choroidal melanoma at McGill University Health Centre from 1997 to 2002. All patients were treated with episcleral iodine-125 (I^{125}) plaque radiotherapy. Plaques were inserted at the tumor site under echographic visualization. All patients had medium-size tumors, except for three. Patients had periodic ophthalmic evaluation at 3 and 6 months post radiation treatment, followed by 6-month intervals thereafter. Patients' mean age was 62 ± 15 years, 16 males and 12 females. Fifty percent of the tumors were located posterior to the equator with significant reduction in size at 12 months post plaque radiotherapy treatment. Significant regression was observed in all the tumor diameters at 5 years post treatment follow-up. Reduction in the depth diameter was significant ($P < 0.01$) in both male and female groups post treatment. There was a 25% ($P < 0.001$) reduction in the medium size of tumors at 5-year follow-up. Tumors located posterior to the equator responded best to I^{125} plaque radiotherapy. Male patients responded better than females to treatment. Medium-size melanoma responded well to plaque radiotherapy.

Key words: choroidal melanoma; I^{125} ; radiotherapy

Introduction

Choroidal melanoma is the most common primary intraocular malignancy and is the only primary ocular tumor that can be fatal in adults.¹ Approximately half of patients die from the disease within 10 to 15 years of enucleation.^{2,3} Despite diagnostic advances and introduction of new treatment modalities, the rate of metastasis has not been substantially re-

duced,² with highest rates of metastasis occurring within the first 5 years after enucleation.¹ Plaque radiotherapy is an important treatment alternative for patients with intraocular tumor. It is the most common form of radiotherapy used for choroidal melanoma.⁴⁻⁶ It offers patients the possibility of eye preservation and possible useful vision.⁷ However it is associated with a 4–18% rate of tumor recurrence.^{6,8-13} In addition, the effects of irradiation on healthy tissues surrounding the radioactive plaque with iodine-125 (I^{125}) and ruthenium-106 (Ru^{106}) have been found to be less damaging to normal ocular structures when compared with other radioactive materials, such as cobalt-60 (Co^{60}).^{14,15}

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TABLE 1. Patients’ Demographic Data and Number of Patients at Each Follow-up Visit

	Total patients [†]		Months after intervention							
	<i>n</i>	Mean age	3	6	12	18	24	36	48	60
Male	16	62 ± 13	16	15	14	13	11	9	4	0
Female	12	63 ± 18	12	11	10	10	9	8	6	3

[†]At the beginning of the study.

Patients and Methods

Clinical records and echographic data of all patients with choroidal melanoma who were treated with I¹²⁵ plaque irradiation along with intra-operative echographic plaque localization at McGill University Health Centre (MUHC) between January 1997 and December 2002 were reviewed. Diagnosis of choroidal melanoma was made prior to the treatment. The Collaborative Ocular Melanoma Study (COMS) protocol was followed for categorization of tumor size. A tumor was classified as small if the height (depth) was less than 2.5 mm and/or base (transverse) diameter was less than or equal to 8 mm. The tumor was considered medium in size if the height was equal to or greater than 2.5 mm and less than or equal to 10.0 mm and/or its base diameter was greater than or equal to 8.1 and less than or equal to 16.0 mm. Large tumors were those of height greater than or equal to 10.1 and/or base diameter greater or equal to 16.1 mm.¹⁶ Tumors were also classified according to their location in the eye. Tumors were considered in location *I* (equatorial), if 50% or more of the diameter was located within the equator area; location *II* (posterior to the equator), if 50% or more of the tumor diameter was posterior to the equator; location *III* (anterior), if more than 50% of the diameter of the tumor was anterior to equator; and location *IV* (close to optic nerve), if it was centered within 5° from the optic nerve. Patients were followed-up at a mean interval of 6 months for 5 years postoperatively and once a year thereafter. Age, sex, eye involved, and tumor diameters (transverse, height, and depth) were recorded prior to and

after plaque radiotherapy. During these follow-up visits; visual acuity, intraocular pressure, slit lamp, and dilated fundal exam were performed for both eyes, in addition to echographic examinations for the involved eye. Complete systemic evaluation, chest X-ray, and liver function tests were performed upon diagnosis and annually thereafter. Forty-eight eyes in 48 patients underwent plaque radiotherapy insertion during the study period. Exclusion criteria were follow-up of less than 3 months post plaque insertion (3 patients), loss of follow-up of 3 or more consecutive visits (1 patient), enucleation (1 patient required enucleation at 3 months post radioactive plaque insertion due to failure of local control and continued growth of the tumor), and Ru¹⁰⁶ plaque insertion (15 patients). Inclusion criteria were at least 3 months follow-up post radioactive plaque insertion and I¹²⁵ plaque radiotherapy use as the treatment modality. A total of 28 patients were included in this study. Student’s *t*-tests, one-way analysis of variance (ANOVA), and polynomial regression analysis were performed with the Prism Instat program.

Results

Twenty-eight eyes of 28 patients, 16 males (57%) and 12 females (43%) with mean age of 62 ± 15 years, were included in the analysis (Table 1). With at least 1 year follow-up in all patients, no metastasis was detected. Retinal detachment was present in nine eyes (32%), additional nonmelanoma tumor was present in four (14%) patients. All patients had medium-size tumors based on the COMS classification,

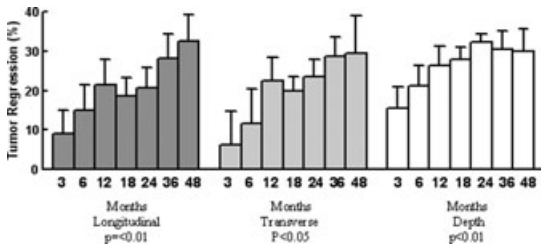


Figure 1. Tumor response to plaque radiotherapy in the male group. The different diameters (longitudinal, transverse, or depth) were measured over a 4-year follow-up period.

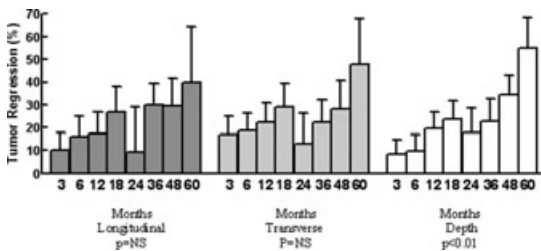


Figure 2. Tumor response to plaque radiotherapy in the female group. The different diameters (longitudinal, transverse or depth) were measured over 5-year follow-up period.

except for three lesions, which were 0.25 mm out of the range, two of these fell in the large-size group and one in the small-size group. Tumors at the equator accounted for 18% (5/28) and anterior to the equator 18% (5/28); 14% (4/28) were near the optic nerve, and 50% (14/28) posterior to the equator. During a mean of 5 years follow-up there was a significant reduction in size of 25% ($P < 0.001$) in the medium-size tumor group, 43% in the small-size, and 8% in the large-size group. However, there the numbers of patients in the latter two groups were too small for detection of statistical significance. Figure 1 shows a reduction of the tumor size—statistically significant in the longitudinal, transverse, and depth diameters—in the male group, while in the female group a significant reduction was observed in the depth diameter only (Fig. 2). Tumors located posterior and anterior to the equator responded well to the plaque radioactive treatment, with 31% and 30% reductions in size respectively at 5 years follow-up.

Discussion

In this study males were predominant; this is consistent with the study of Yanoff and Zimmerman who reported a 6:4 male-to-female ratio in a group of 100 eyes referred to the Armed Forces Institute of Pathology.¹⁷ The male group responded better than the female group in terms of regression in all tumor diameters, while females showed a significant reduction only in depth. However, studies showed similar long-term survival rates in both groups. This supports the view that tumor depth is the main parameter to monitor melanoma growth, or it may implicate some other differences between the two genders that deserve further investigation.

Medium-size tumors responded well to plaque radiotherapy. The other groups were of small sample size, which made comment on the treatment response difficult.

Conclusion

In our study population tumors located posterior to the equator responded best to the treatment. Despite the differences observed in the effects of the plaque radiotherapy on the transverse and longitudinal diameters in the male and female groups, the only common similar response has been shown to be in the depth variable. Taking into consideration the similar long term survival rate, this might be an adequate indicator for monitoring tumor size during follow up. Plaque radiotherapy is an effective treatment in controlling the growth of choroidal melanoma. In total, tumor diameters regressed by 25–54% depending on the variable measured (longitudinal, transverse, or depth).

Future Research

1. Longer follow-up of plaque-irradiated patients is needed to determine the metastasis rate in the long-term post treatment.

2. Despite previous studies indicating that overall survival of patients treated with plaque radiotherapy is comparable to other management techniques,^{18,19} would incorporating intra-operative echographic localization of plaque for choroidal melanoma improve patient care by enabling the delivery of radiation precisely to the tumor?
3. Why did tumors in the male group respond significantly better in more than one diameter compared to the female group? This needs to be investigated in larger study group.
4. Should plaque radiotherapy be used for choroidal melanoma in locations anterior and posterior to equator tumors, since these tumors responded the best of all to the radioactive plaque treatment? Further studies are required before one can recommend this treatment approach as the “standard of care.”

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. Egan, K.M., J.M. Seddon, R.J. Glynn, *et al.* 1988. Epidemiologic aspect of uveal melanoma. *Surv. Ophthalmol.* **32**: 239–251.
2. Jensen, O.A. 1982. Malignant melanomas of the human uvea: 25-year follow-up of cases in Denmark, 1943–1952. *Acta Ophthalmol.* **60**: 161–182.
3. Ravi, I. 1977. Uveal melanoma in Finland: an epidemiological, clinical, histological and prognostic study. *Acta Ophthalmol.* **133**: 3–64.
4. Augsburger, J.J. 1993. Diagnosis and management of posterior uveal tumors. In *Practical Atlas of Retinal Disease and Therapy*. W.R. Freeman, Ed.: 103–124. Raven Press. New York.
5. Shields, J.A., C.L. Shield, P. De Potter, *et al.* 1993. Plaque radiotherapy for uveal melanoma. *Int. Ophthalmol. Clin.* **33**: 129–135.
6. Tabandeh, H., N.A. Chaudhry, T.G. Murry, *et al.* 2000. Intraoperative echographic localization of iodine-¹²⁵ episcleral plaque for brachytherapy of choroidal melanoma. *Am. J. Ophthalmol.* **129**: 199–204.
7. Finger, P.T. 1997. Radiation therapy for choroidal melanoma. *Surv. Ophthalmol.* **42**: 215–232.
8. Char, D.H., J.M. Quivry, J.R. Castro, *et al.* 1993. Helium ions versus iodine¹²⁵ brachytherapy in the management of uveal melanoma. A prospective, randomized, dynamically balanced trial. *Ophthalmology* **100**: 1547–1554.
9. Karlsson, U.L., J.J. Augsburger, J.A. Shields, *et al.* 1989. Recurrence of posterior uveal melanoma after Co⁶⁰ episcleral plaque therapy. *Ophthalmology* **96**: 382–388.
10. Garreston, B.R., D.M. Roberston & J.D. Earle. 1987. Choroidal melanoma treatment with iodine¹²⁵ brachytherapy. *Arch. Ophthalmol.* **105**: 1394–1397.
11. Packer, S., M. Rotman, P. Salantro. 1984. Iodine-¹²⁵ irradiation of choroidal melanoma: clinical experience. *Ophthalmology* **91**: 1700–1708.
12. Gass, J.D.M. 1985. Comparison of prognosis after enucleation vs cobalt⁶⁰ irradiation of melanomas. *Arch. Ophthalmol.* **103**: 916–923.
13. Lommatzsch, P.K. 1983. B-irradiation of choroidal melanoma with Ru¹⁰⁶/Rh¹⁰⁶ application. *Arch. Ophthalmol.* **101**: 713–717.
14. Lommatzsch, P.K. & I.H. Kirsch. 1988. Ru¹⁰⁶/Rh¹⁰⁶ plaque radiotherapy for malignant melanomas of the choroid: with follow-up results more than 5 years. *Doc. Ophthalmol.* **68**: 225–238.
15. Earle, J.D., W.K. Robert, M.R. Dennis, *et al.* 1987. Selection of Iodine¹²⁵ for the collaborative ocular melanoma study. *Arch. Ophthalmol.* **105**: 763–764.
16. Collaborative Ocular Melanoma Group. 1993. Design and methods of clinical trial for rare condition: the collaborative ocular melanoma study. *Controlled Trials* **3**: 362–391.
17. Yanoff, M., Zimmerman L.E. 1967. Histogenesis of malignant melanoma of uvea. II. Relation of uveal nevi to malignant melanomas. *Cancer* **77**: 493–507.
18. Augsburger, J.J., J.W. Gamel, K. Lauritzen, *et al.* 1990. Cobalt⁶⁰ plaque radiotherapy vs enucleation for posterior uveal melanoma. *Am. J. Ophthalmol.* **109**: 585–592.
19. Augsburger, J.J., Z.M. Correa, J. Friere & L.W. Brady. 1998. Long term survival in choroidal and ciliary body melanoma after enucleation versus plaque radiation therapy. *Ophthalmology* **105**: 1670–1678.